GENOME PODS

Elias Crum[1,2], Ruben Taelman[2], Bart Buelens[1], Ruben Verborgh[2], and Gökhan Ertaylan[1]

[1]*Flemish institute for Technological Research (VITO) Mol, Belgium*

[2]*IDLab, Department of Electronics and Information Systems, Ghent University – imec, Belgium*

Scalable Personalized Medicine?

€ Unfortunately, still quite expensive

But why? Sequencing is getting cheap.

Answer: Data privacy is crucial

Genomic data storage is expensive

Question: Any way to fix that?

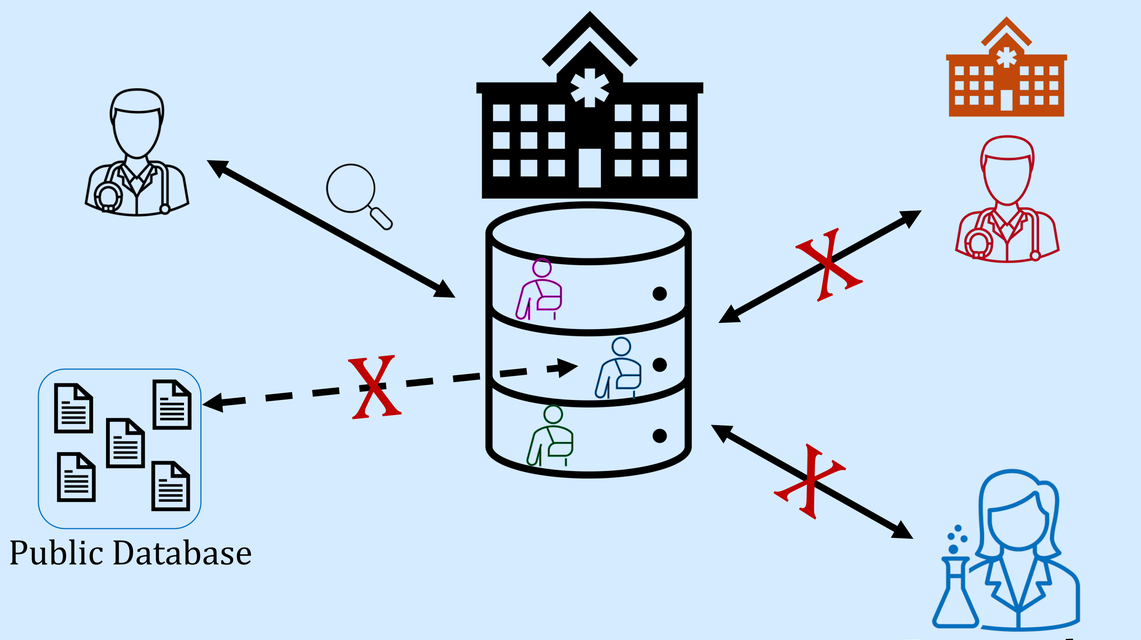
Personal Genome Sequence Data

Variant Call Format (VCF)

Very large (~GBs in size)

Require strict privacy protections

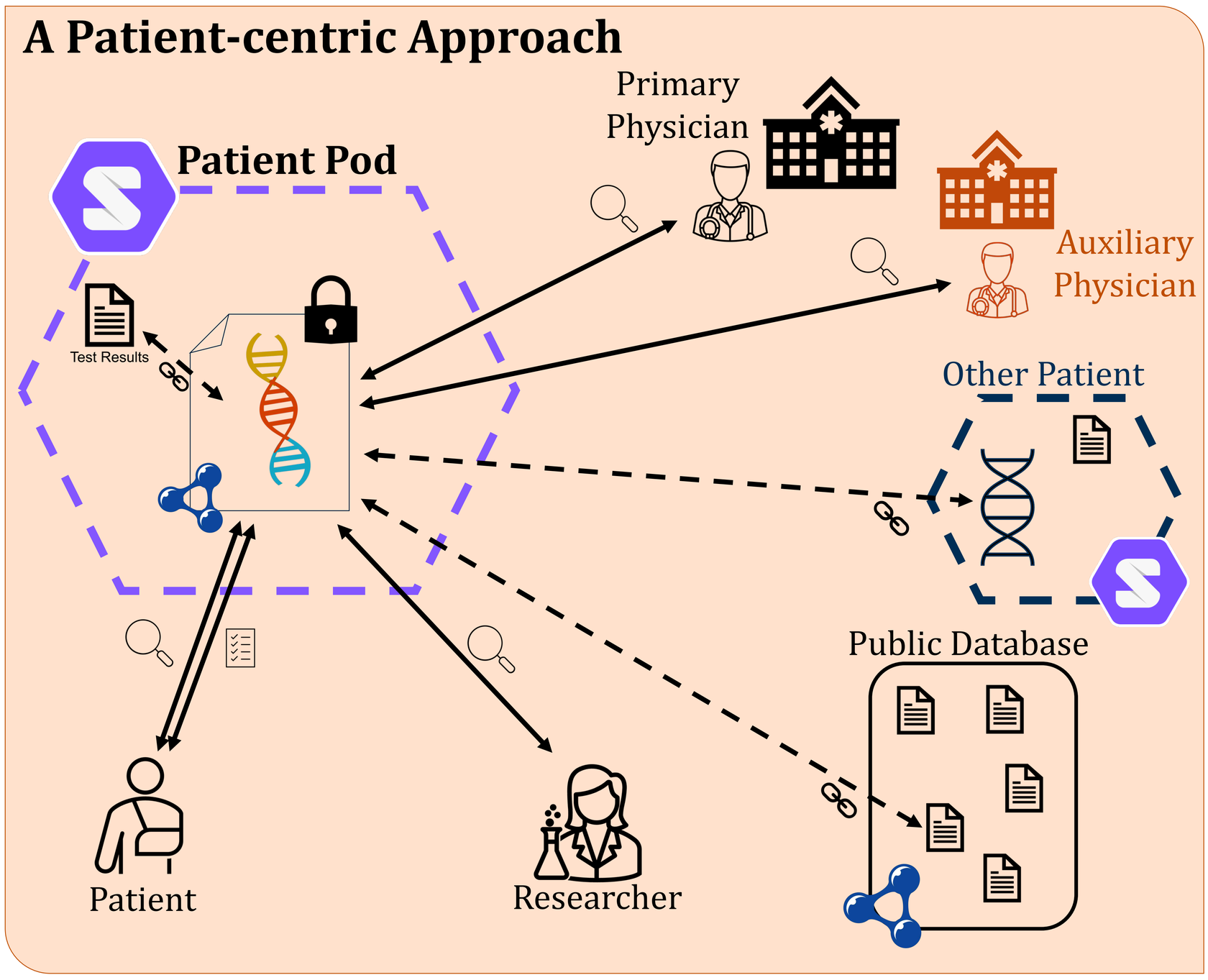
The current institution-centric system



Downstream impacts?

- Reduced data sharing
- Reduced patient transparency
- Increased pertinent data dispersion
- Data duplication

A Patient-centric Approach



Key Challenges

Privacy granularity

- Governance?
- How granular is adequate?

RDF Conversion + Data Linkages

- VCF to RDF Conversion?
- Automation of data linking?

Querying performance

- Many, large data sources
- Algorithms that use indexes?

Engineering Tasks

Genomic data storage in Solid pods

Data privacy management

Data Sharing via the web

Research Pursuits

Represent genomic data as RDF?

Linking genomic and health data?

Single and Many pod Querying?

Funding provided from VITO NV (**UG_PhD_2303_contract**)

vito.be